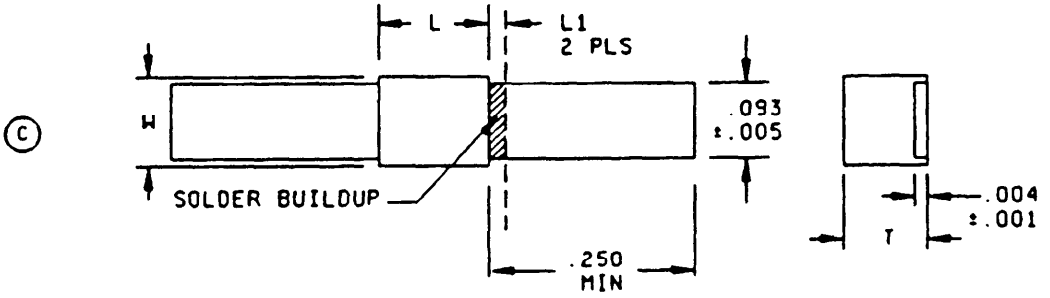
INCH-POUND
MIL-C-55681/5C
13 July 1994
SUPERSEDING
MIL-C-55681/5B
5 April 1985

MILITARY SPECIFICATION SHEET

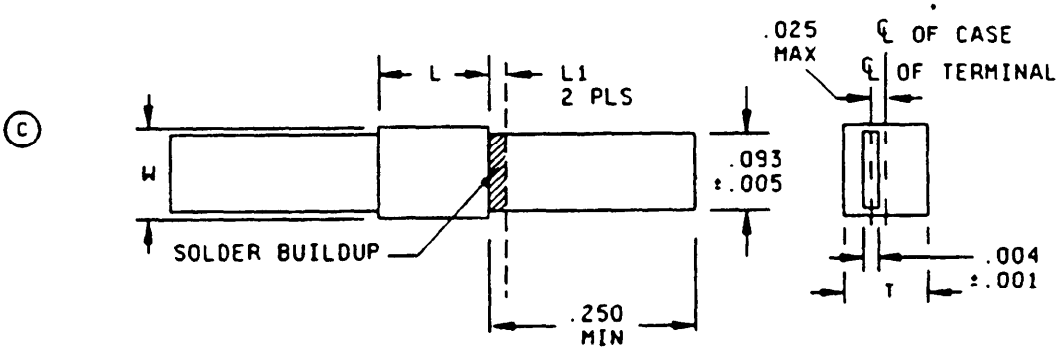
CAPACITORS, CHIP, MULTIPLE LAYER, FIXED, UNENCAPSULATED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY STYLES CDR21, CDR22, CDR23, CDR24, AND CDR25 (HIGH FREQUENCY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55681.



STYLE CDR21



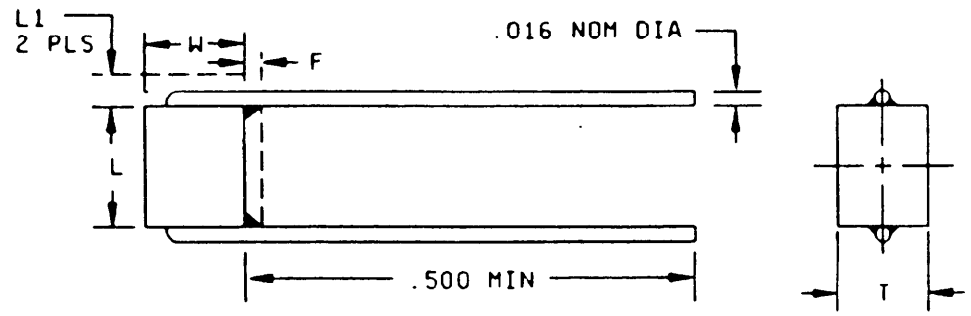
STYLE CDR22

FIGURE 1. Dimensions and configuration.

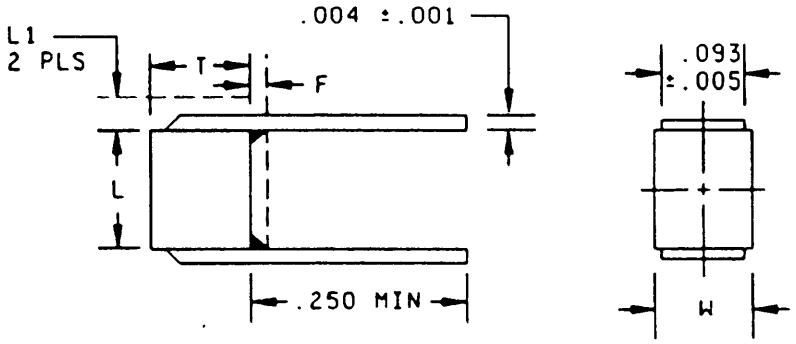
(C) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.



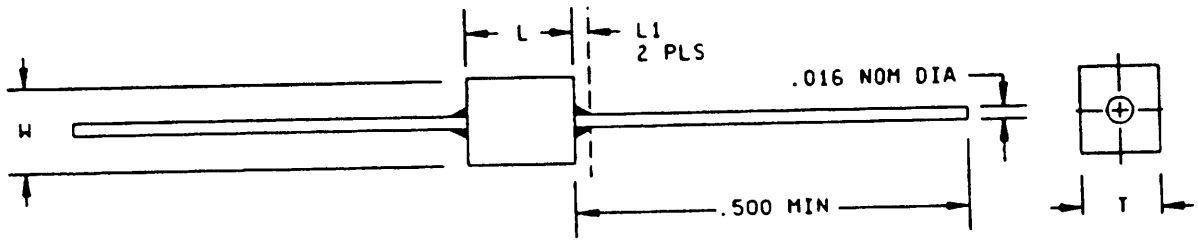
STYLE CDR23



STYLE CDR24

FIGURE 1. Dimensions and configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.



STYLE CDR25

Inches	mm
.001	0.03
.004	0.10
.005	0.13
.015	0.38
.016	0.41
.020	0.51
.025	0.64
.030	0.76
.093	2.36
.100	2.54
.110	2.79
.135	3.43
.250	6.35
.500	12.70

Dimensions						
Style	L	W	T		L1	F
			Min	Max	Max 1/	Max 2/
CDR21	.135 ±.015	.110 ±.015	.030	.100	.020	---
CDR22	.135 ±.015	.110 ±.015	.030	.100	.020	---
CDR23	.135 ±.015	.110 ±.015	.030	.100	.020	.020
CDR24	.135 ±.015	.110 ±.015	.030	.100	.020	.020
CDR25	.135 ±.015	.110 ±.015	.030	.100	.020	---

1/ Solder buildup over each lead.
2/ Solder fill.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Case type: Multi-layer, unencapsulated, monolithic.

Material: Ceramic.

Terminals: Termination finish S or T. Silver leads, when applicable, are attached with solder having a melting temperature of greater than 200°C.

Inspection conditions: All visual examinations shall be performed using a stereo microscope having a minimum magnification of 20 power.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

(C) Dissipation factor: In accordance with MIL-C-55681.

Insulation resistance:

At +25°C ±5°C: 10^6 megohms, minimum (for 0.1 pF through 470 pF).

10^5 megohms, minimum (for 510 pF through 5100 pF).

At +125°C ±5°C: 10^5 megohms, minimum (for 0.1 pF through 470 pF).

10^4 megohms, minimum (for 510 pF through 5100 pF).

(C) Equivalent series resistance: In accordance with MIL-C-55681.

Terminal strength: In accordance with MIL-C-55681.

Marking: In accordance with MIL-C-55681.

NOTE: These capacitors are intended for use in RF solid-state circuitry up to and including 25 gigahertz, and cannot be replaced in the field.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

TABLE I. Styles CDR21, CDR22, CDR23, CDR24, and CDR25 capacitor characteristics.

PIN 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			volts, dc
CDR2-B-0R1EB--	0.1	B	BG, BP	500
CDR2-B-0R2EB--	0.2	B	"	"
CDR2-B-0R3E---	0.3	B,C	"	"
CDR2-B-0R4E---	0.4	B,C	"	"
CDR2-B-0R5E---	0.5	B,C,D	"	"
CDR2-B-0R6E---	0.6	"	"	"
CDR2-B-0R7E---	0.7	"	"	"
CDR2-B-0R8E---	0.8	"	"	"
CDR2-B-0R9E---	0.9	"	"	"
CDR2-B-1R0E---	1.0	"	"	"
CDR2-B-1R1E---	1.1	"	"	"
CDR2-B-1R2E---	1.2	"	"	"
CDR2-B-1R3E---	1.3	"	"	"
CDR2-B-1R4E---	1.4	"	"	"
CDR2-B-1R5E---	1.5	"	"	"
CDR2-B-1R6E---	1.6	"	"	"
CDR2-B-1R7E---	1.7	"	"	"
CDR2-B-1R8E---	1.8	"	"	"
CDR2-B-1R9E---	1.9	"	"	"
CDR2-B-2R0E---	2.0	"	"	"
CDR2-B-2R1E---	2.1	"	"	"
CDR2-B-2R2E---	2.2	"	"	"
CDR2-B-2R4E---	2.4	"	"	"
CDR2-B-2R7E---	2.7	"	"	"
CDR2-B-3R0E---	3.0	"	"	"
CDR2-B-3R3E---	3.3	"	"	"
CDR2-B-3R6E---	3.6	"	"	"
CDR2-B-3R9E---	3.9	"	"	"
CDR2-B-4R3E---	4.3	"	"	"
CDR2-B-4R7E---	4.7	"	"	"
CDR2-B-5R1E---	5.1	"	"	"
CDR2-B-5R6E---	5.6	"	"	"
CDR2-B-6R2E---	6.2	"	"	"
CDR2-B-6R8E---	6.8	B,C,J,K,M	"	"
CDR2-B-7R5E---	7.5	"	"	"
CDR2-B-8R2E---	8.2	"	"	"
CDR2-B-9R1E---	9.1	"	"	"
CDR2-B-100E---	10	F,G,J,K,M	"	"
CDR2-B-110E---	11	"	"	"
CDR2-B-120E---	12	"	"	"
CDR2-B-130E---	13	"	"	"
CDR2-B-150E---	15	"	"	"
CDR2-B-160E---	16	"	"	"
CDR2-B-180E---	18	"	"	"
CDR2-B-200E---	20	"	"	"
CDR2-B-220E---	22	"	"	"
CDR2-B-240E---	24	"	"	"
CDR2-B-270E---	27	"	"	"
CDR2-B-300E---	30	"	"	"
CDR2-B-330E---	33	"	"	"
CDR2-B-360E---	36	"	"	"
CDR2-B-390E---	39	"	"	"
CDR2-B-430E---	43	"	"	"
CDR2-B-470E---	47	"	"	"
CDR2-B-510E---	51	"	"	"
CDR2-B-560E---	56	"	"	"
CDR2-B-620E---	62	"	"	"
CDR2-B-680E---	68	"	"	"
CDR2-B-750E---	75	"	"	"

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

TABLE I. Styles CDR21, CDR22, CDR23, CDR24, and CDR25 capacitor characteristics - Continued.

PIN 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			volts, dc
CDR2-B-820E---	82	F,G,J,K,M	BG, BP	500
CDR2-B-910E---	91	"	"	"
CDR2-B-101E---	100	"	"	300
CDR2-B-111D---	110	"	"	"
CDR2-B-121D---	120	"	"	"
CDR2-B-131D---	130	"	"	"
CDR2-B-151D---	150	"	"	"
CDR2-B-161D---	160	"	"	"
CDR2-B-181D---	180	"	"	"
CDR2-B-201D---	200	"	"	200
CDR2-B-221C---	220	"	"	"
CDR2-B-241C---	240	"	"	"
CDR2-B-271C---	270	"	"	"
CDR2-B-301C---	300	"	"	"
CDR2-B-331C---	330	"	"	"
CDR2-B-361C---	360	"	"	"
CDR2-B-391C---	390	"	"	"
CDR2-B-431C---	430	"	"	"
CDR2-B-471C---	470	"	"	100
CDR2-B-511B---	510	"	"	"
CDR2-B-561B---	560	"	"	"
CDR2-B-621B---	620	"	"	50
CDR2-B-681A---	680	"	"	"
CDR2-B-751A---	750	"	"	"
CDR2-B-821A---	820	"	"	"
CDR2-B-911A---	910	"	"	"
CDR2-B-102A---	1000	"	BP	"
CDR2-B-112A---	1100	"	"	"
CDR2-B-122A---	1200	"	"	"
CDR2-B-132A---	1300	"	"	"
CDR2-B-152A---	1500	"	"	"
CDR2-B-162A---	1600	"	"	"
CDR2-B-182A---	1800	"	"	"
CDR2-B-202A---	2000	"	"	"
CDR2-B-222A---	2200	"	"	"
CDR2-B-242A---	2400	"	"	"
CDR2-B-272A---	2700	"	"	"
CDR2-B-302A---	3000	"	"	"
CDR2-B-332A---	3300	"	"	"
CDR2-B-362A---	3600	"	"	"
CDR2-B-392A---	3900	"	"	"
CDR2-B-432A---	4300	"	"	"
CDR2-B-472A---	4700	"	"	"
CDR2-B-502A---	5000	"	"	"
CDR2-B-512A---	5100	"	"	"

1/ Complete PIN will include additional symbols to indicate style, voltage-temperature limits, capacitance tolerance (where applicable), termination finish ("T" for styles CDR21, CDR22, and CDR24, and "S" for styles CDR23 and CDR25), and failure rate level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - AS, MC, OS, SH
Air Force - 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-1838-02)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.